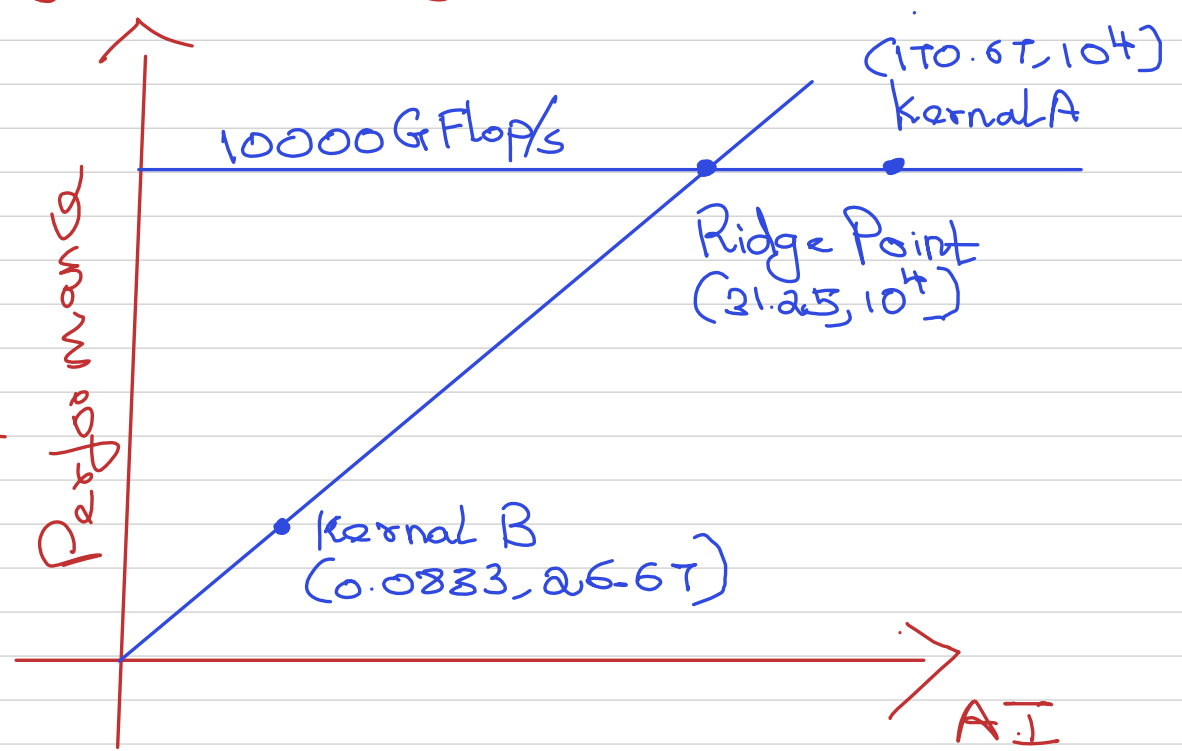


1. Roofline Diagram.



$$\text{Ridge point} = \left(\frac{10^4}{320}, 10^4 \right)$$

2. Kernel A.

Multiplying two F.P. 1024 x 1024

$$\text{Flops} = 2N^3 = 2 \times 1024^3 \approx 2.147 \text{ GFlops}$$

$$\text{Bytes Transferred} = 3 \times 1024^2 \times 4 \approx 12.58 \text{ MB}$$

$$\text{AI} = \frac{\text{Flops}}{\text{Bytes}} = 170.67 \text{ Flop/byte}$$

Attainable Performance

$$P = \min(10,000, 320 \times 170.67) = 10,000 \text{ GFlops/s}$$

3. Kernel B

$$N = 4, 194, 304$$

Element wise Addition

$$C[i] = A[i] + B[i]$$

$$\text{Flops} \approx 0.00419 \text{ GFLOPs}$$

$$\text{Bytes Transferred} = 3 \times N \times 4 \approx 50.33 \text{ MB}$$

$$\text{AI} = \frac{1}{12} \approx 0.0833 \text{ Flop/Byte}$$

Attainable Performance

$$P = \min(10,000, 26.67 \text{ GF/s}) = 26.67 \text{ GFlop/s}$$

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	kernel A	kernel B
Flops	2.14 GFlops	0.00419 GFlops
Bytes	12.58 MB	50.33 MB
AI	170.67 Flop/byte	0.0833 F/byte
Attainable Performance	10,000 GFlop/s	26.61 GF/s
Bottleneck	Compute Bound	Memory Bound
Best Architectural Change	Increasing peak compute	Increasing Memory B.W